UNIFORM SHEAFY TATE DOMAINS THAT ARE NOT STABLY UNIFORM

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Abstract. Buzzard–Verberkmoes and Mihara show that a stably uniform Tate Huber ring is sheafy. We construct two uniform sheafy Tate domains which are not stably uniform, answering Question 7 of Kedlaya's *Nonarchimedean Scottish Book*. In the first example a rational localisation is nonreduced, whilst in the second the nonuniform rational localisation is an integral domain. The two main results are due to ChatGPT 5.6 Sol. Both constructions, including their sheafiness and the rational localisations witnessing failure of stable uniformity, have been formalised in Lean 4.

1. Introduction

The point at issue is the behaviour of rational localisations. The examples of Buzzard–Verberkmoes and Mihara show that a uniform Tate Huber ring can lose uniformity on a rational chart and, separately, that uniformity alone does not imply sheafiness [4, 17]. They leave open the possibility of a uniform sheafy ring which is not stably uniform. Hansen and Kedlaya ask this explicitly [7, Remark 3.16]; it is also Question 7 of Kedlaya's *Nonarchimedean Scottish Book* [18]. The answer is no.

ChatGPT 5.6 Sol found both examples and supplied the proofs. The Lean formalisation was carried out by Claude Code. We have aimed to cite related work that may have informed this process; see remark 1.3. Section 9 gives a fuller account, and appendix A records the formal definitions and theorem statements.

We begin with the "finite-jet pullback". This is a nonnoetherian uniform integral domain for which a distinguished rational localisation is nonreduced, and hence is no longer uniform. Let k be a complete discretely valued nonarchimedean field, with valuation ring k°, uniformizer ϖ, and residue field $\widetilde{k}$. Put

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$
$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ means $k^\circ\langle W, V\rangle/(WV-1)$. There are natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$. Define

$$A_0 = B_0 \times_{D_0} C_0, \qquad A = A_0[1/\varpi],$$

and endow A with the topology defined by the ring of definition A_0.

The main result is the following.

Theorem 1.1. *The ring A is a complete uniform nonnoetherian Tate k-algebra and an integral domain, with $A^\circ = A_0$. The Huber pair (A, A°) is sheafy. However,*

$$A\left\langle \frac{W}{\varpi}\right\rangle \cong k\langle X, Q\rangle/(Q^2), \qquad X = \frac{W}{\varpi},$$

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so A is not stably uniform.

The same rational chart gives a negative answer to Problem 24 of the *Nonarchimedean Scottish Book* and an affirmative answer to the Tate case of Problem 28; see section 7. These two consequences have also been formalised in Lean after specialising to $k = \mathbb{Q}((t))$.

The proof uses Milnor squares of complete Tate rings. Recall that a cartesian square of rings with a surjective leg is called a Milnor square [16, Section 2]; the complete Tate setting is used by Kerz–Saito–Tamme in their work on analytic K-theory [13, Section 3].

Definition 1.2. A commutative square of complete Tate k-algebras

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological k-vector spaces

$$0 \longrightarrow R \longrightarrow B \oplus C \xrightarrow{(b,c)\mapsto \bar{b}-\bar{c}} D \longrightarrow 0$$

is strict exact, and $C \to D$ is a strict surjection.

After rotating the square if necessary, the initial Milnor square is strict by the nonarchimedean open mapping theorem [13, Lemma 3.1]. Our main input is that, for this finite-jet square, the strict exact row persists after completed rational localisation and is compatible with refinement. This transfers sheafiness from B, C, D to A. The second example shows that the failure of stable uniformity need not come from a nonreduced localisation.

Remark 1.3 (Relation with earlier work)**.** It is difficult to understand exactly what sources an AI agent such as ChatGPT may have used in doing this work. In the interest of giving appropriate credit, we have listed here closely related work that may have been used in the development of the results here.

Buzzard–Verberkmoes construct a uniform affinoid ring with a nonuniform rational localisation and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [4, Propositions 17 and 18]. Mihara gives further examples of both phenomena [17]. These examples do not give a uniform sheafy ring which is not stably uniform.

Ben-Bassat–Kremnizer prove strict two-term resolutions for Weierstrass and Laurent localisations, and reduce a general rational localisation to these cases [2, Lemmas 5.13–5.14]. Bambozzi–Kremnizer formulate rational localisation in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. The lattice estimates in lemma 5.1 are the form needed to show that rational localisation preserves the finite-jet Milnor square. The resulting uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 8 is closer to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara. Its additional feature is that rational localisation can be calculated in a noetherian finite-variable subring.

2. Conventions and notation

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [10], and use Wedhorn's account for strong noetherianity and sheafiness [20, Proposition and Definition 6.36, Section 8.1]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [4], and the quotient presentation of a rational localisation is also discussed by Hansen–Kedlaya [7].

- *Tate topology.* Throughout, E denotes a complete Tate k-algebra. In the discretely valued setting of this paper, this means that E contains a topologically nilpotent unit and admits an open bounded k°-subalgebra E_0 such that
$$E = E_0[1/\varpi], \qquad \{\varpi^n E_0\}_{n\geq 0}$$
is a basis of neighbourhoods of zero. Such an E_0 is called a *ring of definition.* We always choose it closed, hence ϖ-adically complete. A subset of E is bounded if it is contained in $\varpi^{-r}E_0$ for some $r \geq 0$. If M is a complete topological k-vector space, an open bounded k°-submodule $M_0 \subset M$ will be called a lattice.
- *Uniformity and noetherianity.* We write E° for the ring of power-bounded elements. The ring E is *uniform* if E° is bounded, and *strongly noetherian* if $E\langle Z_1, \dots, Z_s\rangle$ is noetherian for every $s \geq 0$. A Tate ring is *stably uniform* if every rational localisation is uniform.
- *Huber pairs and sheafiness.* A *ring of integral elements* is an open subring $E^+ \subseteq E^\circ$ which is integrally closed in E. We call the Huber pair (E, E^+) *sheafy* when its structure presheaf, valued in complete topological rings, satisfies the sheaf condition. Thus, for a finite rational cover $U = \bigcup_i U_i$, restriction identifies $\mathcal{O}(U)$ homeomorphically with the subspace of $\prod_i \mathcal{O}(U_i)$ consisting of tuples whose restrictions to every $U_i \cap U_j$ agree.
- *Strictness.* We use the following bounded-denominator form of strictness. Let $u : M \to N$ be a continuous k-linear map and choose lattices $M_0 \subset M$ and $N_0 \subset N$. Then u is strict if and only if there exists $a \geq 0$ such that
$$\varpi^a\big(u(M) \cap N_0\big) \subset u(M_0). \tag{1}$$
For a surjection this becomes $\varpi^a N_0 \subset u(M_0)$. For an injection it says that the lattice induced from the target is bounded in the source. This is the criterion used by Buzzard–Verberkmoes [4, Lemma 2]. We use the term *strict Milnor square* in the sense of definition 1.2; the underlying algebraic terminology is standard [16, 13].
- *Rational localisation.* We take every rational datum to have at least one numerator; an empty list may be padded by $f_1 = 0$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in E, so the ideal generated by $g, f_1, \dots, f_m$ is open. Since E is a Tate k-algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of ϖ, we may assume that all entries lie in E_0.

  The rational localisation E_α is the separated ϖ-adic completion of $E[1/g]$ for the ring of definition generated by
$$E_0\left[\frac{f_1}{g}, \dots, \frac{f_m}{g}\right].$$
It is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Put
$$T_E = E\langle T_1, \dots, T_m\rangle = E_0\langle T_1, \dots, T_m\rangle[1/\varpi], \qquad r_i = gT_i - f_i,$$
and
$$I_{E,\alpha} := (r_1, \dots, r_m) = \mathrm{im}\left(T_E^m \xrightarrow{d_{1,E}} T_E\right) \subset T_E, \qquad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i(gT_i - f_i), \tag{2}$$
Then
$$E_\alpha \cong T_E/\overline{I_{E,\alpha}}. \tag{3}$$

Here the bar is necessary until the ideal has been shown to be closed. This is the standard quotient presentation of rational localisation; see [10, (1.2), Proposition 1.3, Lemma 1.5] and [7, Definition 3.6].

3. The finite-jet ring

In this section we identify the finite-jet algebra with a concrete subring of C and prove that it is a complete uniform domain. Recall that

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$
$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ denotes $k^\circ\langle W, V\rangle/(WV-1)$. The natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$ define

$$A_0 = B_0 \times_{D_0} C_0.$$

We write

$$A = A_0[1/\varpi], \quad L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

Each of these Tate rings carries the topology defined by its subscript-zero ring.

The quotient map $C_0 \to D_0$ has a continuous k°-linear section

$$\sigma : D_0 \longrightarrow C_0, \qquad \sigma(f_0 + Qf_1) = f_0 + Qf_1,$$

obtained by viewing the unique representative of Q-degree at most one as an element of C_0. Consequently

$$0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0 \tag{4}$$

is exact, and A_0 is ϖ-adically complete. After inverting ϖ we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

In fact, the chosen rings of definition already make the integral row exact, so no powers of ϖ are lost in the defining square.

The map $B_0 \to D_0$ is injective. Projection to C_0 therefore identifies A_0 with the subring

$$A_0 = \left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W\rangle,\ h \in C_0 \right\}. \tag{5}$$

Equivalently, A_0 consists of the restricted series

$$\sum_{(a,b)\in S} c_{a,b} W^a Q^b, \qquad S = \{(a,b) \in \mathbb{Z}\times\mathbb{N} : b \le 1 \Rightarrow a \ge 0\}, \tag{6}$$

with $c_{a,b} \in k^\circ$. Here restricted means that, for every n, only finitely many coefficients are nonzero modulo ϖ^n. The set S is closed under addition, so this is a subring of C_0. The same description with coefficients in k gives $A \subset C$.

Equip A, B, C, D with the coefficientwise Gauss norms whose unit balls are the displayed rings of definition. Write

$$\iota_B : A \to B, \qquad \iota_C : A \to C, \qquad \rho_B : B \to D, \qquad \rho_C : C \to D$$

for the structural maps, and give A the restricted Gauss norm from C. For $a = f_0 + Qf_1 + Q^2 h$, the coefficient formulas give

$$\|\iota_C(a)\|_C = \|a\|_A, \qquad \|\iota_B(a)\|_B \le \|a\|_A, \qquad \|\rho_B(b)\|_D \le \|b\|_B, \qquad \|\rho_C(c)\|_D \le \|c\|_C. \tag{7}$$

Thus every structural map is norm-nonincreasing.

Proposition 3.1. *The ring A is a complete uniform nonnoetherian Tate k-algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

Proof. The Gauss norm on $L = k\langle W, W^{-1}\rangle$ is multiplicative, and the restricted Gauss norm on $C = L\langle Q\rangle$ is therefore multiplicative as well. In particular, C is a domain, and hence so is its subring A. The conditions on the constant and linear Q-coefficients which define A are closed conditions in C, so A is complete. The image of ϖ in A is a topologically nilpotent unit, and therefore A is a Tate ring.

The coefficient description of A_0 shows that it is exactly the unit ball for the restricted Gauss norm on A. Since this norm is multiplicative,

$$\|a^n\| = \|a\|^n.$$

It follows that a is power-bounded if and only if $\|a\| \leq 1$. Thus $A^\circ = A_0$, and the unit ball is bounded, so A is uniform.

For nonnoetherianity, let $J = \ker(\iota_B) = Q^2C \subset A$. If J were generated by $a_1, \dots, a_r$, write $b_i \in L$ for the coefficient of Q^2 in a_i. For every $\ell \in L$ we could write

$$Q^2\ell = \sum_i x_i a_i, \qquad x_i \in A,$$

and write c_i for the constant coefficient of x_i as a series in Q. By equation (5), we have $c_i \in k\langle W\rangle$. Since $a_i \in Q^2C$, its coefficients in degrees 0 and 1 vanish. The coefficient of Q^2 in $x_i a_i$ is therefore exactly $c_i b_i$. Taking the coefficient of Q^2 in the displayed equality gives

$$\ell = \sum_i c_i b_i, \qquad c_i \in k\langle W\rangle.$$

Thus $b_1, \dots, b_r$ would generate L as a $k\langle W\rangle$-module. This is impossible. Indeed, it would make W^{-1} integral over $k\langle W\rangle$. Multiplying a monic relation for W^{-1} of degree n by W^n would put 1 in the proper ideal $(W) \subset k\langle W\rangle$. Thus J is not finitely generated, and A is not noetherian. □

4. A nonuniform rational localisation

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum

$$\alpha_{\text{in}} = (W; \varpi).$$

The ideal (ϖ, W) is open, since ϖ is a unit in A. Let

$$A_{\text{in}} = A\left\langle \frac{W}{\varpi} \right\rangle$$

be the corresponding rational localisation on $|W| \leq |\varpi|$, and write $X = W/\varpi$. Since ϖ is already invertible in A, the algebraic localisation does not change the ring. More explicitly, A_{in} is obtained by taking the separated ϖ-adic completion of $A_0[X] \subset A$ and then inverting ϖ.

Proposition 4.1. *There is a canonical isomorphism of complete topological rings*

$$A_{\text{in}} \cong k\langle X, Q\rangle/(Q^2), \qquad W \longmapsto \varpi X, \quad Q \longmapsto Q.$$

Proof. Put

$$T_0 = k^\circ\langle X, Q\rangle/(Q^2), \qquad T = T_0[1/\varpi].$$

Let $\rho : A \to A_{\text{in}}$ be the canonical map. We first show that

$$Q^2C \subseteq \ker(\rho). \tag{8}$$

Take $y \in Q^2C$, and choose $r \geq 0$ such that $\varpi^r y \in Q^2C_0$. For every $n \geq 0$, the coefficient description of A_0 gives

$$W^{-n}\varpi^r y \in A_0 :$$

multiplication by W^{-n} only introduces negative powers of W in terms of Q-degree at least two. In $A_0[X]$, where $W = \varpi X$, we therefore have

$$\varpi^r y = W^n(W^{-n}\varpi^r y) = \varpi^n X^n(W^{-n}\varpi^r y) \in \varpi^n A_0[X]$$

for every n. Hence $\varpi^r y$ maps to zero in the separated ϖ-adic completion of $A_0[X]$. Since ϖ is invertible in A_{in}, this proves (8).

We now construct the isomorphism. Recall that the map $\iota_B : A \to B$ from the defining pullback is given by

$$f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(W) + Qf_1(W);$$

it simply discards the terms divisible by Q^2. The substitution $W \mapsto \varpi X$, $Q \mapsto Q$ defines a continuous homomorphism $B \to T$. Their composite is

$$\theta : A \longrightarrow T, \qquad f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(\varpi X) + Qf_1(\varpi X).$$

It sends A_0 into T_0 and sends W/ϖ to X. It therefore sends $A_0[X]$ into T_0 and extends, first to the separated completions and then after inverting ϖ, to a continuous homomorphism

$$\Phi : A_{\mathrm{in}} \longrightarrow T.$$

Set

$$\overline{X} = \rho(W)/\varpi, \qquad \overline{Q} = \rho(Q).$$

Both elements are power-bounded: they belong to the completed ring of definition coming from $A_0[X]$. Moreover, $\overline{Q}^2 = 0$ by (8). Evaluation at $\overline{X}$ therefore gives a continuous homomorphism

$$\Psi : T \longrightarrow A_{\mathrm{in}}, \qquad f + Qg \longmapsto f(\overline{X}) + g(\overline{X})\overline{Q}.$$

It remains to check the two composites. Write $a = f_0(W) + Qf_1(W) + y \in A$, with $y \in Q^2C$. Then

$$\Psi\Phi(\rho(a)) = f_0(\varpi\overline{X}) + \overline{Q}f_1(\varpi\overline{X}) = \rho\big(f_0(W) + Qf_1(W)\big) = \rho(a),$$

where the last equality uses (8). The image of A is dense in A_{in}, so continuity gives $\Psi\Phi = 1$. Conversely, $\Phi\Psi$ fixes k, X and Q, and hence is the identity on the dense subring $k[X,Q]/(Q^2)$ of T. It follows by continuity that $\Phi\Psi = 1$. Thus Φ and Ψ are inverse continuous homomorphisms. □

Proposition 4.2. *The ring A is not stably uniform.*

Proof. In $T = k\langle X, Q\rangle/(Q^2)$, every element of the line kQ is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition T_0: for every $r \geq 0$,

$$\varpi^r\big(\varpi^{-(r+1)}Q\big) = \varpi^{-1}Q \notin T_0.$$

Thus T is not uniform. By proposition 4.1, a rational localisation of the uniform ring A is nonuniform. □

5. Localisation of the Milnor square

In this section we prove that rational localisation preserves the defining pullback. We first record the strict Koszul estimate needed for the proof.

5.1. **Koszul complexes for rational localisations.** Let E be one of B, C, D, with its coefficientwise Gauss norm and the displayed unit ball E_0. Let $f_1, \dots, f_m, g \in E_0$ generate the unit ideal in E. Put

$$T_E = E\langle T_1, \dots, T_m\rangle, \qquad T_{E,0} = E_0\langle T_1, \dots, T_m\rangle,$$

and set $r_i = gT_i - f_i$.

We give T_E the topology for which $\{\varpi^n T_{E,0}\}_{n\geq 0}$ is a basis of neighbourhoods of zero. The term $\bigwedge^q T_E^m$ has the finite-module topology defined by the lattice $\bigwedge^q T_{E,0}^m$; equivalently, in the standard exterior basis it has the finite product topology. Images have the subspace topology. Thus a differential is strict when it is open onto its image, or equivalently when its coimage is homeomorphic to its image.

Applied to $E \in \{B, C, D\}$, the strict Koszul regularity theorem of Bambozzi–Kremnizer [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13] gives the exactness, strictness, and closed-image assertions below. We include the direct proof followed in the formalisation because the two estimates relative to the displayed lattices are not stated there and are needed below.

Lemma 5.1. *The Koszul complex*

$$0 \longrightarrow \bigwedge^m T_E^m \longrightarrow \cdots \longrightarrow \bigwedge^2 T_E^m \xrightarrow{d_{2,E}} T_E^m \xrightarrow{d_{1,E}} T_E$$

is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if

$$I_E = (r_1, \dots, r_m) = \operatorname{im}(d_{1,E}) \subset T_E, \qquad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i r_i,$$

then I_E is closed. Moreover, there exists $h_E \in \mathbb{Z}_{\geq 0}$ such that

$$\varpi^{h_E}(I_E \cap T_{E,0}) \subset d_{1,E}(T_{E,0}^m). \tag{9}$$

There also exists $z_E \in \mathbb{Z}_{\geq 0}$ such that

$$\varpi^{z_E}\big(\ker(d_{1,E}) \cap T_{E,0}^m\big) \subset d_{2,E}\big(\textstyle\bigwedge^2 T_{E,0}^m\big). \tag{10}$$

Proof. Over the polynomial ring $E[T_1, \dots, T_m]$, the sequence $gT_i - f_i$ has Koszul complex exact in positive degrees. The map

$$E[T_1, \dots, T_m] \longrightarrow T_E$$

is flat: this is the usual identification of T_E with the ϖ-adic completion of $E_0[T_1, \dots, T_m]$, followed by inverting ϖ. Tensoring the polynomial Koszul complex with T_E, and using the standard base-change identification of its finite free terms, proves positive-degree exactness. Notice that this does not assert exactness in degree zero.

Each differential is continuous because its coordinates are finite sums of multiplication maps. The image of $d_{1,E}$ is closed by the noetherian closed-ideal theorem. In higher degree, exactness identifies an image with the kernel of the next continuous differential, and hence that image is closed. The nonarchimedean open mapping theorem therefore shows that, for each differential $d_{q,E}$, there is a constant $C_{q,E} \geq 1$ such that every $y \in \operatorname{im}(d_{q,E})$ has a preimage x satisfying

$$d_{q,E}(x) = y, \qquad \|x\| \leq C_{q,E}\|y\|.$$

This also proves that each differential is strict onto its image.

Choose $h_E \geq 0$ such that $|\varpi|^{h_E} C_{1,E} \leq 1$. If $x \in I_E \cap T_{E,0}$, choose $u \in T_E^m$ with

$$d_{1,E}(u) = x, \qquad \|u\| \leq C_{1,E}\|x\| \leq C_{1,E}.$$

Then $\varpi^{h_E}u \in T^m_{E,0}$ and $d_{1,E}(\varpi^{h_E}u) = \varpi^{h_E}x$, which proves (9). Similarly, choose $z_E \geq 0$ such that $|\varpi|^{z_E}C_{2,E} \leq 1$. If $u \in \ker(d_{1,E}) \cap T^m_{E,0}$, choose v with $d_{2,E}(v) = u$ and $\|v\| \leq C_{2,E}\|u\|$. Then $\varpi^{z_E}v \in \bigwedge^2 T^m_{E,0}$, proving (10). $\square$

5.2. **The defining ideals under pullback.** Fix a rational datum

$$\alpha = (f_1, \dots, f_m; g)$$

in A. Multiplying all entries by a common power of ϖ, we assume $f_1, \dots, f_m, g \in A_0$. Put

$$r_i = gT_i - f_i \in T_A,$$

and use the same notation for its image in T_B, T_C, and T_D. For $E \in \{A, B, C, D\}$ put

$$I_E = (r_1, \dots, r_m) = \operatorname{im}(d_{1,E}) \subset T_E.$$

The exact integral row (4) and the coefficientwise section of $C_0 \to D_0$ give an exact sequence

$$0 \longrightarrow T_{A,0} \longrightarrow T_{B,0} \oplus T_{C,0} \longrightarrow T_{D,0} \longrightarrow 0. \tag{11}$$

The map $T_{C,0} \to T_{D,0}$ is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to A_0, and the lifted coefficients still tend to zero ϖ-adically.

Lemma 5.2. *The ideal I_A is closed in T_A, and the canonical map*

$$I_A \longrightarrow I_B \times_{I_D} I_C$$

is an isomorphism. Give each I_E the subspace topology inherited from T_E, and give $I_B \oplus I_C$ the max norm. Then the sequence of topological k-vector spaces

$$0 \longrightarrow I_A \longrightarrow I_B \oplus I_C \longrightarrow I_D \longrightarrow 0 \tag{12}$$

is strict exact. Here the maps are

$$x \longmapsto (x_B, x_C), \qquad (x_B, x_C) \longmapsto (x_B)_D - (x_C)_D.$$

Proof. For $E = B, C, D$, let $C_{1,E}$ be as in the proof of lemma 5.1, and let $C_{2,D}$ be the corresponding constant for $d_{2,D}$. Take $(x_B, x_C) \in I_B \times_{I_D} I_C$, and choose $u_B \in T^m_B$ and $u_C \in T^m_C$ such that

$$d_{1,B}(u_B) = x_B, \qquad \|u_B\| \leq C_{1,B}\|x_B\|,$$
$$d_{1,C}(u_C) = x_C, \qquad \|u_C\| \leq C_{1,C}\|x_C\|.$$

In T^m_D the difference

$$w = (u_B)_D - (u_C)_D$$

belongs to $\ker(d_{1,D})$. Choose $v_D \in \bigwedge^2 T^m_D$ such that

$$d_{2,D}(v_D) = w, \qquad \|v_D\| \leq C_{2,D}\|w\|.$$

The norm-preserving coefficientwise section $T_D \to T_C$ induces a norm-preserving section on second exterior powers; use it to obtain $v_C \in \bigwedge^2 T^m_C$ mapping to v_D. Replacing u_C by

$$u'_C = u_C + d_{2,C}(v_C)$$

does not change $d_{1,C}(u_C)$, and the images of u_B and u'_C in T^m_D now agree. Exactness of (11), applied coefficientwise, gives $u_A \in T^m_A$ mapping to this pair. Then $d_{1,A}(u_A)$ maps to (x_B, x_C).

The three displayed inequalities, continuity of $d_{2,C}$, and the norm-preserving coefficientwise section give a constant C, independent of (x_B, x_C), for which $x_A = d_{1,A}(u_A)$ satisfies

$$\|x_A\| \leq C \max\{\|x_B\|, \|x_C\|\}.$$

The map $T_A \to T_B \oplus T_C$ is injective, so this proves both the algebraic pullback assertion and the boundedness of its inverse.

It remains to see that I_A is closed. The ideals I_B and I_C are closed by lemma 5.1. If x lies in the closure of I_A, its images lie in I_B and I_C. The pullback isomorphism just proved gives $x_A \in I_A$ with the same two images as x. Injectivity of $T_A \to T_B \oplus T_C$ gives $x = x_A$, as required.

Finally, for $y \in I_D$ choose $u_D \in T_D^m$ such that

$$d_{1,D}(u_D) = y, \qquad \|u_D\| \leq C_{1,D}\|y\|.$$

Apply the norm-preserving section $T_D^m \to T_C^m$ and then $d_{1,C}$. This gives $x_C \in I_C$ mapping to y, with $\|x_C\|$ bounded by a fixed multiple of $\|y\|$. Thus $I_B \oplus I_C \to I_D$ is a strict surjection. Its kernel is I_A by the pullback assertion, and the preceding bound for the inverse $I_B \times_{I_D} I_C \to I_A$ makes the inclusion on the left strict. This proves (12). ∎

Proposition 5.3. *For every rational datum α in A, the canonical sequence*

$$0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0 \tag{13}$$

is strict exact, and $C_\alpha \to D_\alpha$ is a strict surjection. Equivalently,

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

The coefficient maps used in these identifications commute with rational restriction.

Proof. The ideals I_B, I_C, I_D are closed by lemma 5.1, and I_A is closed by lemma 5.2. Hence (3) gives

$$E_\alpha = T_E/I_E \qquad (E = A, B, C, D).$$

We first prove algebraic exactness. If a representative $p \in T_A$ maps into both I_B and I_C, the pullback assertion in lemma 5.2 gives an element of I_A with the same two images. Injectivity of $T_A \to T_B \oplus T_C$ then shows that $p \in I_A$. This gives injectivity on the left. Conversely, take compatible classes in T_B/I_B and T_C/I_C and choose representatives b and c. Their defect in T_D belongs to I_D. Lift this defect to an element of I_C using the strict surjection in (12), and add it to c. The corrected representatives agree in T_D, so (11) lifts them to T_A. This proves (13).

For the topology on the left, let $x \in A_\alpha$ and put

$$M = \max\{\|x_B\|, \|x_C\|\}.$$

Choose representatives $b \in T_B$ and $c \in T_C$ whose norms are arbitrarily close to the quotient norms of x_B and x_C. Their defect lies in I_D. The norm estimate for $I_C \to I_D$ proved in lemma 5.2 gives a constant C_0 and a correction $y \in I_C$ mapping to this defect such that

$$\|y\| \leq C_0\|b_D - c_D\|.$$

Thus b and $c + y$ agree in T_D and lift to $p \in T_A$. The max-norm identity for the ambient pullback gives, after letting the error in the representatives tend to zero,

$$\|x\| \leq CM$$

for a constant C depending only on the datum. The two quotient maps are norm-nonincreasing, so this estimate says precisely that $A_\alpha \to B_\alpha \oplus C_\alpha$ is a topological embedding.

The coefficientwise section $T_D \to T_C$ shows directly that $C_\alpha \to D_\alpha$ is surjective. More precisely, choose a representative in T_D whose norm is arbitrarily close to the quotient norm of a given class in D_α, and apply the norm-preserving section. Its class in C_α has no larger norm. Thus the surjection is open; together with continuity, this says that it is strict.

All maps used above come from the coefficient maps $A \to B, C$ and $B, C \to D$. More generally, a homomorphism $E \to E'$ carrying a rational datum α to α' induces a coefficientwise map $T_E \to T_{E'}$ which sends the relations for α to those for α'. The pullback identifications therefore commute with these maps. Applying this to rational restriction, and then using presentation independence and transitivity, proves the asserted compatibility under refinement. □

6. Sheafiness

Theorem 6.1. *The Huber pair (A, A°) is sheafy.*

Proof. Give B, C, D their maximal rings of integral elements. The max/Gauss estimates (7) show that every structural map is norm-nonincreasing. Consequently it sends bounded subsets to bounded subsets; in particular, if x is power-bounded, then the set of powers of its image is the image of the bounded set $\{x^n : n \geq 0\}$. The structural maps therefore send power-bounded elements to power-bounded elements and define morphisms of Huber pairs.

Each of B, C, D is an affinoid k-algebra and is therefore sheafy by Huber's theorem [10, Theorem 2.2]. Let $U = \bigcup_i U_i$ in $\mathrm{Spa}(A, A^\circ)$. Push the datum for U and the data for the U_i to B, C, D. If a section over U restricts to zero on every U_i, its images over B and C vanish by separatedness there. Injectivity of the localised Milnor row from proposition 5.3 then shows that the original section is zero.

Now take a matching family over the U_i. Its images glue over B and C. The two glued sections have the same image over D, since their restrictions agree on the pushed cover and the presheaf for D is separated. Exactness of the localised Milnor row over U gives a section over A, and injectivity of the corresponding rows over the U_i shows that it restricts to the given family.

We must also check the topological part of the sheaf condition. The image of

$$\mathcal{O}(U) \longrightarrow \prod_i \mathcal{O}(U_i)$$

is exactly the subspace of tuples whose two restrictions to every $U_i \cap U_j$ agree. This subspace is closed: it is the common zero locus of the continuous difference maps to $\mathcal{O}(U_i \cap U_j)$. The restriction map is continuous and injective, so the nonarchimedean closed-range theorem, using the topologically nilpotent unit of A, shows that its inverse on this image is continuous. Hence the restriction map is a topological embedding. This proves the sheaf condition on finite rational covers. The sheaf-on-a-rational-basis theorem [19, Tag 009O] then gives the all-open topological sheaf condition for (A, A°). □

Proof of theorem 1.1. Completeness, uniformity, nonnoetherianity, and the domain property are proposition 3.1; sheafiness is theorem 6.1; and failure of stable uniformity is proposition 4.2. □

7. Problems 24 and 28

Problem 24 of the *Nonarchimedean Scottish Book* asks whether the map of rings underlying a rational localisation of Tate Huber pairs must be flat. Problem 28 asks whether a non-zero-divisor which generates a closed ideal can restrict to zero on a rational subspace [18, Problems 24 and 28]. The same rational datum, together with multiplicativity of the Gauss norm, answers both questions. We give the argument over the field k fixed in the introduction. The Lean formalisation treats the special case $k = F((t))$, which is enough for the two existence questions.

Proposition 7.1. *Let*

$$U = \{x \in \mathrm{Spa}(A, A^\circ) : |W(x)| \leq |\varpi(x)|\}.$$

Then U is nonempty, the rational localisation

$$A \longrightarrow \mathcal{O}(U) = A\left\langle \frac{W}{\varpi} \right\rangle$$

is not flat, and multiplication by Q^2 is a strict inclusion on A even though Q^2 maps to zero in $\mathcal{O}(U)$.

Proof. The composite

$$A \longrightarrow B = k\langle W, Q\rangle/(Q^2) \longrightarrow k, \qquad f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(0),$$

is norm-nonincreasing. It therefore sends A° into k°, and pulling back the norm valuation on k gives a continuous valuation belonging to $\operatorname{Spa}(A, A^\circ)$. At this point W has value zero and ϖ has nonzero value, so the point lies in U.

Since the restricted Gauss norm on A is multiplicative and $\|Q\| = 1$, for every $a \in A$ we have

$$\|Q^2a\| = \|a\|.$$

Thus multiplication by Q^2 is an isometry. Its map onto its image, with the subspace topology, is consequently a homeomorphism, so the inclusion is strict. Moreover, an isometric image of the complete space A is closed in A. Since $\|Q^2\| = 1$, the element Q^2 is nonzero; multiplication by it is therefore injective because A is a domain.

On the other hand, the calculation (8) says directly that the canonical map to the completed rational localisation kills Q^2. The isomorphism of proposition 4.1 identifies this localisation with the nonzero ring

$$A \longrightarrow k\langle X, Q\rangle/(Q^2), \qquad W \longmapsto \varpi X, \quad Q \longmapsto Q.$$

If the canonical map were flat, the non-zero-divisor Q^2 would still act injectively on $\mathcal{O}(U)$; equivalently, tensoring the injection

$$0 \longrightarrow A \xrightarrow{\cdot Q^2} A$$

with $\mathcal{O}(U)$ would make multiplication by the image of Q^2 injective. Its image is zero, however, so this multiplication map is zero on the nonzero ring $\mathcal{O}(U)$, a contradiction. □

In Lean, `presheafValue (chartDatum F)` is the completed rational localisation, not the ordinary algebraic localisation. The formalised witness checks that the datum is rational, that U is nonempty, that its section ring is nonzero, and that multiplication by Q^2 is continuous, injective, strict, and has closed image before proving that Q^2 restricts to zero. Taking $F = \mathbb{Q}$ gives the required examples. This does not address the perfectoid part of Problem 28.

8. A second example

The finite-jet construction above loses uniformity because separated completion creates a nilpotent. We now give a second construction in which the bad rational localisation is an integral domain. The loss of uniformity is instead caused by an unbounded family of nonnilpotent power-bounded elements.

Theorem 8.1. *There is a complete Tate k-algebra $\mathcal{A}$ with the following properties.*

(1) *$\mathcal{A}$ is a uniform, nonnoetherian integral domain and $\mathcal{A}^\circ = \mathcal{A}_0$ for the ring of definition displayed below.*
(2) *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.*
(3) *The rational localisation*

$$\mathcal{B} = \mathcal{A}\left\langle \frac{W}{\varpi} \right\rangle$$

is an integral domain but is not uniform.

Consequently $\mathcal{A}$ is not stably uniform. In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localisation.

8.1. **The weighted-parity algebra.** We write $\mathbb{N} = \{0, 1, 2, \ldots\}$ and put $I = \mathbb{N}_{>0}$. For a finite-support multi-index $\nu = (\nu_n)_{n \in I} \in \mathbb{N}^{(I)}$, put

$$\omega(\nu) = \sum_{\substack{n \geq 1 \\ \nu_n \text{ odd}}} n. \tag{14}$$

Consider the submonoid

$$S = \left\{ (a, \nu) \in \mathbb{N} \times \mathbb{N}^{(I)} : a \geq \omega(\nu) \right\}. \tag{15}$$

The inequality

$$\omega(\nu + \mu) \leq \omega(\nu) + \omega(\mu) \tag{16}$$

shows that S is closed under addition. It is locally finite: a fixed element of S has only finitely many decompositions as a sum of two elements of S.

Let $k[S]$ be the monoid algebra with basis $W^a U^\nu$, where $U^\nu = \prod_n U_n^{\nu_n}$. Give it the Gauss norm

$$\left\| \sum_{(a,\nu) \in S} c_{a,\nu} W^a U^\nu \right\|_G = \sup_{a,\nu} |c_{a,\nu}|. \tag{17}$$

Its completion is

$$\mathcal{A} = \left\{ \sum_{(a,\nu) \in S} c_{a,\nu} W^a U^\nu : c_{a,\nu} \in k, \quad c_{a,\nu} \longrightarrow 0 \right\}, \tag{18}$$

where convergence to zero is over the discrete index set S. Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on $k[S]$ extends multiplication continuously to $\mathcal{A}$. Put

$$\mathcal{A}_0 = \left\{ \sum_{(a,\nu) \in S} c_{a,\nu} W^a U^\nu \in \mathcal{A} : c_{a,\nu} \in k^\circ \right\}. \tag{19}$$

Then $\mathcal{A}_0$ is the closed unit ball for (17), it is ϖ-adically complete, and $\mathcal{A} = \mathcal{A}_0[1/\varpi]$.

Put

$$Y_n = W^n U_n, \qquad Z_n = U_n^2,$$

and, for $N \in \mathbb{N}$, let $\mathcal{A}^{(N)} \subset \mathcal{A}$ be the closed subring supported on monomials involving only $W, U_1, \ldots, U_N$. Thus $\mathcal{A}^{(0)} = k\langle W \rangle$. Let $\mathcal{T}^{(N)} \subset \mathcal{A}^{(N)}$ be the subring in which every U_n has even exponent.

Lemma 8.2 (Finite-variable subrings)**.** *For every $N \in \mathbb{N}$, substitution $Z_n \mapsto U_n^2$ gives a norm-preserving ring isomorphism*

$$k\langle W, Z_1, \ldots, Z_N \rangle \xrightarrow{\sim} \mathcal{T}^{(N)}.$$

For $N = 0$ this is the evident identification of both sides with $k\langle W \rangle$. Every $f \in \mathcal{A}^{(N)}$ can be written

$$f = \sum_{\epsilon \in \{0,1\}^N} f_\epsilon Y^\epsilon, \qquad f_\epsilon \in \mathcal{T}^{(N)}, \qquad Y^\epsilon = \prod_{n=1}^{N} Y_n^{\epsilon_n}. \tag{20}$$

Consequently $\mathcal{A}^{(N)}$ is finite over $\mathcal{T}^{(N)}$ and is a strongly noetherian Tate ring. The inclusions $\mathcal{A}^{(N)} \to \mathcal{A}^{(M)} \to \mathcal{A}$ for $N \le M$ preserve the norm. Moreover, for every $f \in \mathcal{A}$ and $\ell \ge 0$ there are N and $f_N \in \mathcal{A}^{(N)}$ such that

$$\|f - f_N\|_G \le |\varpi|^\ell \|f\|_G.$$

In particular,

$$\mathcal{A} = \widehat{\bigcup_{N\ge 0} \mathcal{A}^{(N)}}. \tag{21}$$

Proof. On even exponents, halving the U_n-coordinates identifies the support monoid with that of the Tate algebra $k\langle W, Z_1, \ldots, Z_N\rangle$; the coefficient map and its inverse preserve the Gauss norm. Write each exponent uniquely as $\nu_n = 2q_n + \epsilon_n$, with $\epsilon_n \in \{0, 1\}$. Since $\omega(\nu) = \sum_{\epsilon_n=1} n$, every allowed monomial factors as

$$W^a U^\nu = W^{a-\omega(\nu)} \prod_{n=1}^N Z_n^{q_n} \prod_{n=1}^N Y_n^{\epsilon_n}.$$

Grouping the monomials with the same tuple $(\epsilon_1, \ldots, \epsilon_N)$ gives (20). The finitely many Y^ϵ therefore generate $\mathcal{A}^{(N)}$ over $\mathcal{T}^{(N)}$. The same argument after adjoining any finite number of Tate variables gives strong noetherianity. The inclusions are coefficientwise and hence isometric. Finally, approximate f to the required precision by a finite sum of monomials and choose N so that only $U_1, \ldots, U_N$ occur in that sum. This gives the stated estimate and (21). □

Proposition 8.3. *The ring $\mathcal{A}$ is a complete uniform Tate k-algebra and an integral domain, with*

$$\mathcal{A}^\circ = \mathcal{A}_0.$$

Proof. The Gauss norm on the countable restricted Tate algebra $k\langle W, U_1, U_2, \ldots\rangle$ is multiplicative. To see this, note first that the norm of a nonzero restricted series is attained. For each of two such series choose, among the norm-attaining exponents, the lexicographically largest one. In the coefficient of the sum of these two exponents, the product of the chosen coefficients strictly dominates every other convolution term. The ultrametric inequality therefore gives

$$\|fg\|_G = \|f\|_G \|g\|_G.$$

This is the countable restricted-series argument used in the formalisation, whose Lean implementation uses William Coram's restricted-power-series code [6]. The support algebra $\mathcal{A}$ is a closed isometric subring of this Tate algebra, so its Gauss norm is multiplicative and it is a domain.

If $x \in \mathcal{A}_0$, all powers of x remain in $\mathcal{A}_0$. If $x \notin \mathcal{A}_0$, multiplicativity gives $\|x^m\|_G = \|x\|_G^m$, which is unbounded. Hence $\mathcal{A}^\circ = \mathcal{A}_0$. Completeness and the Tate property follow from (18) and the topologically nilpotent unit ϖ. □

Proposition 8.4. *The ring $\mathcal{A}$ is not noetherian.*

Proof. For $m \ge 1$, let

$$I_m = (Z_1, \ldots, Z_m) \subset \mathcal{A}.$$

Define a norm-nonincreasing homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T\rangle,$$

by retaining only the coefficients of the monomials U_{m+1}^{2q} and sending that monomial to T^q. The support condition shows directly that this is multiplicative: a product can contribute such a monomial only when both factors contribute monomials of the same form. Thus ψ_m sends W and every Y_j to zero, sends Z_{m+1} to T, and sends every other Z_j to zero. It kills I_m but not Z_{m+1}. Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots,$$

so $\mathcal{A}$ is not noetherian. □

8.2. **Small perturbations of rational data.**

Lemma 8.5 (Small perturbation lemma)**.** *Let (E, E^+) be a complete Tate Huber pair and let $\alpha = (f_1, \ldots, f_m; g)$ be a rational datum in E. There is a neighbourhood V of zero in E such that, whenever*

$$g' - g,\ f_1' - f_1, \ldots, f_m' - f_m \in V,$$

the tuple $\alpha' = (f_1', \ldots, f_m'; g')$ is a rational datum and α and α' define the same rational subset of $\operatorname{Spa}(E, E^+)$*. Their completed rational localisations are canonically bicontinuously isomorphic over E.*

Proof. Apply [9, Lemma 3.10] after including g among the numerators; see also [12, Remark 2.4.7]. The assertion about completed rational localisations follows from their universal property [10, Proposition 1.3]. □

8.3. **Rational localisations defined over finitely many variables.** Fix $N \in \mathbb{N}$. Let $\mu = (\mu_n)_{n>N}$ be a finitely supported family of nonnegative integers and put

$$e_\mu = W^{\omega(\mu)} U^\mu, \tag{22}$$

where μ is extended by zero for $n \leq N$. Every $f \in \mathcal{A}$ has a unique convergent expression

$$f = \sum_\mu f_\mu e_\mu, \qquad f_\mu \in \mathcal{A}^{(N)}, \qquad \|f_\mu\|_G \longrightarrow 0, \tag{23}$$

and

$$\|f\|_G = \sup_\mu \|f_\mu\|_G.$$

Multiplication is determined by

$$e_\mu e_\lambda = W^{\omega(\mu)+\omega(\lambda)-\omega(\mu+\lambda)} e_{\mu+\lambda}. \tag{24}$$

The map

$$\rho_N : \mathcal{A} \longrightarrow \mathcal{A}^{(N)}, \qquad \sum_\mu f_\mu e_\mu \longmapsto f_0, \tag{25}$$

is a norm-nonincreasing ring retraction.

Proposition 8.6 (Coefficientwise localisation)**.** *Let α be a rational datum whose entries lie in $\mathcal{A}^{(N)}$, and put $P = (\mathcal{A}^{(N)})_\alpha$. There is a bicontinuous ring isomorphism*

$$\mathcal{A}_\alpha \xrightarrow{\sim} \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}. \tag{26}$$

The ring on the right has the supremum norm, and its multiplication is given by (24), with W replaced by its image in P. If a rational refinement is also represented by data in $\mathcal{A}^{(N)}$ and $P \to P'$ is the corresponding restriction map, then the induced restriction sends

$$\sum_\mu p_\mu e_\mu \longmapsto \sum_\mu (p_\mu|_{P'}) e_\mu. \tag{27}$$

Proof. Apply the canonical map $\mathcal{A}^{(N)} \to P$ to every coefficient in (23). This gives a continuous homomorphism

$$\Phi_0 : \mathcal{A} \longrightarrow \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}.$$

In the target, the image of g is a unit and each f_i/g is of norm at most 1, by the defining property of P. Thus Φ_0 extends continuously to $\mathcal{A}[1/g]$ for the rational-localisation topology, and then to the completion:

$$\Phi : \mathcal{A}_\alpha \longrightarrow \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}.$$

In the other direction, the inclusion $\mathcal{A}^{(N)} \to \mathcal{A}$ gives a continuous homomorphism $P \to \mathcal{A}_\alpha$. The family $(e_\mu)_\mu$ is bounded in $\mathcal{A}_\alpha$. Apply the homomorphism to every coefficient. If $\|p_\mu\| \to 0$, the series

$$\sum_\mu p_\mu e_\mu$$

converges in $\mathcal{A}_\alpha$. The resulting map Ψ is continuous, and (24) shows that it is multiplicative.

The composite $\Phi\Psi$ fixes every term pe_μ, and hence every finite sum of such terms. These finite sums are dense. The other composite fixes the dense image of $\mathcal{A}[1/g]$. Continuity therefore gives $\Phi\Psi = 1$ and $\Psi\Phi = 1$, proving (26).

If $P \to P'$ comes from a rational refinement, all the maps just used commute with this restriction map. On a finite sum, restriction simply applies $P \to P'$ to each coefficient; continuity gives (27) for every convergent sum. □

Corollary 8.7 (Finite-variable presentation)**.** *Every rational localisation of $\mathcal{A}$ is bicontinuously isomorphic to a ring of the form*

$$\left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}, \tag{28}$$

where P is a rational localisation of one of the strongly noetherian Tate rings $\mathcal{A}^{(N)}$.

Proof. Start with a rational datum $\alpha = (f_1, \ldots, f_m; g)$ in $\mathcal{A}$. Choose a scalar $t \in k^\times$ with $0 < |t| < 1$ and small enough that $tg, tf_1, \ldots, tf_m$ lie in $\mathcal{A}_0$. Multiplying the datum by t does not change the rational subset or its localisation, so we may assume that $g, f_1, \ldots, f_m$ lie in $\mathcal{A}_0$. Choose a neighbourhood V as in lemma 8.5. By (21), for some N there are $g', f_1', \ldots, f_m' \in \mathcal{A}^{(N)} \cap \mathcal{A}_0$ such that

$$g' - g,\ f_1' - f_1, \ldots, f_m' - f_m \in V.$$

By lemma 8.5, the datum $\alpha' = (f_1', \ldots, f_m'; g')$ gives the same rational subset and the same completed localisation. It is rational in $\mathcal{A}$, so applying ρ_N to a Bezout identity for its entries shows that it is already a rational datum in $\mathcal{A}^{(N)}$. Now apply proposition 8.6. □

8.4. **A reduced nonuniform rational chart.** The ideal (W, ϖ) is open because ϖ is a unit of $\mathcal{A}$. Define the complete k°-algebra

$$\mathcal{B}_0 = \left\{ \sum_{\nu \in \mathbb{N}^{(I)},\ d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \longrightarrow 0 \right\}, \tag{29}$$

with norm

$$\left\|\sum b_{d,\nu} X^d U^\nu\right\|_{\mathrm{in}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|. \tag{30}$$

The subadditivity (16) shows that $\mathcal{B}_0$ is a ring. Put $\mathcal{B} = \mathcal{B}_0[1/\varpi]$.

Proposition 8.8. *There is a bicontinuous ring isomorphism*

$$\mathcal{A}\left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}, \qquad W \longmapsto \varpi X. \tag{31}$$

Proof. The datum $(W; \varpi)$ lies in $\mathcal{A}^{(0)} = k\langle W \rangle$. Let

$$P = \mathcal{A}^{(0)}\langle T \rangle / \overline{(\varpi T - W)}$$

be its separated completed quotient. Evaluation at $W = \varpi X$ and $T = X$ gives a continuous map $P \to k\langle X \rangle$. Conversely, evaluation of $k\langle X \rangle$ at the image of T gives a continuous map in the other direction. The two maps are inverse by polynomial density, and the norm estimates in the two universal properties show that the isomorphism is isometric.

Apply proposition 8.6 with $N = 0$. It describes the rational chart as the ring of sums

$$\sum_\nu p_\nu e_\nu, \qquad p_\nu \in P, \qquad \|p_\nu\| \longrightarrow 0.$$

Under $P \cong k\langle X \rangle$, this is precisely the model (29). Its sup norm is (30), because the term indexed by ν is multiplied by $\varpi^{\omega(\nu)}$. The map from $\mathcal{A}^{(0)}$ sends W to ϖX. □

Proposition 8.9. *The ring $\mathcal{B}$ is an integral domain but is not uniform.*

Proof. The coefficient description gives a homomorphism

$$\mathcal{B} \longrightarrow k\langle X \rangle[[U_1, U_2, \ldots]], \qquad \sum_\nu p_\nu e_\nu \longmapsto \sum_\nu (\varpi X)^{\omega(\nu)} p_\nu(X) U^\nu.$$

It is a homomorphism by (24). If its image is zero, then

$$(\varpi X)^{\omega(\nu)} p_\nu(X) = 0$$

for every ν. Since $k\langle X \rangle$ is a domain, every p_ν is zero. The map is therefore injective. Its target is a domain, and hence so is $\mathcal{B}$.

Inside $\mathcal{A}$ write $Y_n = W^n U_n$ and put

$$T_n = \varpi^{-n} Y_n = X^n U_n \in \mathcal{B}.$$

Formula (30) gives

$$\|T_n\|_{\mathrm{in}} = |\varpi|^{-n}, \qquad \|T_n^2\|_{\mathrm{in}} = 1.$$

The even powers of T_n have norm at most 1, and the odd powers have norm at most $\|T_n\|_{\mathrm{in}}$. Thus every T_n is power-bounded. The family $(T_n)_{n \geq 1}$ is not bounded, since its norms $|\varpi|^{-n}$ are unbounded. Hence $\mathcal{B}^\circ$ is unbounded and $\mathcal{B}$ is not uniform. □

8.5. **Sheafiness.**

Theorem 8.10. *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.*

Proof. We first work with a finite rational cover $U = \bigcup_i U_i$. Choose one N large enough to approximate the datum for U and the data for all the U_i in $\mathcal{A}^{(N)}$. The small perturbation lemma does not change the rational domains, and proposition 8.6 gives compatible identifications

$$\mathcal{O}_X(U) \cong \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P,\ \|p_\mu\| \to 0 \right\},$$

$$\mathcal{O}_X(U_i) \cong \left\{ \sum_\mu p_{i,\mu} e_\mu : p_{i,\mu} \in P_i,\ \|p_{i,\mu}\| \to 0 \right\},$$

with coefficientwise restriction maps. Here P is a rational localisation of the strongly noetherian Tate ring $\mathcal{A}^{(N)}$, and the P_i are rational localisations corresponding to the U_i. Let V and V_i be the rational subsets of

$$\mathrm{Spa}\big(\mathcal{A}^{(N)}, (\mathcal{A}^{(N)})^\circ\big)$$

defined by the data giving P and P_i. Then $V = \bigcup_i V_i$. Indeed, if $v \in V$, the composite $v \circ \rho_N$ is a point of $\mathrm{Spa}(\mathcal{A}, \mathcal{A}^\circ)$, since ρ_N sends power-bounded elements to power-bounded elements. For data in $\mathcal{A}^{(N)}$, membership in the corresponding rational subset is unchanged by this pullback because ρ_N is a retraction. Thus $v \circ \rho_N$ belongs to U, and hence to some U_i, which gives $v \in V_i$. We may therefore apply sheafiness of the strongly noetherian ring $\mathcal{A}^{(N)}$ to the cover $V = \bigcup_i V_i$.

Let $(x_i)_i$ be a matching family and write $x_i = \sum_\mu x_{i,\mu} e_\mu$. Since $\mathcal{A}^{(N)}$ is sheafy, the matching family $(x_{i,\mu})_i$ glues to a unique $q_\mu \in P$ for every μ. It remains to check that $\|q_\mu\| \to 0$. The restrictions of the q_μ to every P_i tend to zero because the coefficients of each x_i do. The product restriction map

$$P \longrightarrow \prod_i P_i$$

is a topological embedding, so it reflects convergence to zero. Hence $q_\mu \to 0$, and $\sum_\mu q_\mu e_\mu$ is the required section over U. The same coefficientwise argument proves separation.

The image of the product restriction map for $\mathcal{A}$ is therefore the subspace of tuples whose restrictions to every pairwise intersection agree. This subspace is closed: it is the common zero locus of the continuous difference maps. The restriction map is continuous and injective. The nonarchimedean closed-range theorem over the Tate ring $\mathcal{A}$ shows that its inverse on this image is continuous, so restriction is a topological embedding. We have proved the topological sheaf condition on finite rational covers. The general rational-basis theorem upgrades this to the all-open structure presheaf, and its plus-ring independence gives the same conclusion for every ring of integral elements. In particular, $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy. □

Proof of theorem 8.1. The uniform domain assertion is proposition 8.3, and nonnoetherianity is proposition 8.4. Sheafiness is theorem 8.10. Finally, propositions 8.8 and 8.9 identify the rational chart $|W| \le |\varpi|$ as a nonuniform integral domain. □

Remark 8.11 (Relation with the first construction)**.** The finite-jet and weighted-parity examples use different mechanisms. In the finite-jet ring, all terms of Q-degree at least two become infinitely ϖ-divisible, and the bad chart acquires a nilpotent. In the weighted-parity ring, the data defining any finite rational calculation can be approximated inside some strongly noetherian $\mathcal{A}^{(N)}$, after which localisation and restriction are computed coefficient by coefficient in the remaining variables. Its bad chart remains a domain, and nonuniformity is witnessed by the unbounded family $(T_n)_{n\ge1}$ of nonnilpotent power-bounded elements. The use of an infinite monomial support and of unbounded power-bounded directions is in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the coefficientwise calculation over the rings $\mathcal{A}^{(N)}$ is the additional feature that makes sheafiness accessible here.

9. How the examples were found

The authors had for some time been interested in finding a counterexample to Problem 7 of the *Nonarchimedean Scottish Book*. Our attempts had centred on the example of [4, Section 4.5, Proposition 17], which is uniform but not stably uniform. We believe that this example is in fact sheafy, but we have not found a proof.

In January 2026 we began to experiment with AI tools, initially ChatGPT 5.2 Pro. Buzzard–Verberkmoes reduce exactness for a two-member Laurent cover to a boundedness condition on the relevant rings of definition [4, Lemma 2]. For a uniform ring this follows from boundedness of the power-bounded elements [4, Lemma 3 and Corollary 4]. Their reduction from general rational covers to Laurent covers is given in [4, Lemma 8 and the proof of Theorem 7]. For sheafiness these checks must also be made after rational localisation. Their Theorem 7 cannot be applied directly to Proposition 17, since one of the rational localisations in that example is not uniform. Our plan was instead to ask ChatGPT to write down many explicit rational covers and check the required exactness directly for each one, in the hope that a pattern, and eventually a proof, would emerge.

After a couple of weeks ChatGPT 5.2 Pro claimed to have a complete proof. The authors then began the slow process of understanding this proof, only to discover, with the help of ChatGPT 5.3, that one of its arguments used the following assertion:

If $\phi : A \to B$ is a continuous homomorphism of Hausdorff topological abelian groups and A is complete, then $\phi(A)$ is closed in B.

This assertion is false, and its use completely broke the purported proof.

Motivated by this, we began in parallel to use Claude Code to formalise the parts of the theory of adic spaces needed here, including the theorem that strongly noetherian Tate rings are sheafy. The aim was a practical one: future AI-generated proofs could be formalised before the human authors had spent days understanding an argument containing a fatal error near its end.

We then returned to writing down covers and looking for a pattern. This continued through ChatGPT 5.3 and subsequent versions. When ChatGPT 5.6 Sol became available, we asked, almost in passing, whether it could prove the result we wanted or *find a new example*. It produced the examples studied here. By then the Lean development was far enough advanced that we could use Claude Code to formalise the arguments quickly in Lean and check that they were correct.

Appendix A. Lean formalisation and verification

This appendix records the Lean formalisation of the two examples. The finite-jet declarations are taken from AINTLIB commit `b007a4f3d`, and the weighted-parity declarations from commit `090a28921`. The comparison with Mathlib's sheaf predicate is taken from commit `d92f96504`, and the consequences for Scottish Book Problems 24 and 28 from commit `d5e626ccc` [1]. For each example Lean proves the Tate structure, uniformity, domain and nonnoetherianity statements, sheafiness, and failure of stable uniformity, together with the rational chart calculation used in the proof.

Neither example snapshot was available from the public AINTLIB repository at the time of writing. The interactive version of the paper includes each declaration used below, extracted from the appropriate commit. The later sheaf-comparison and Scottish Book commits are public, and their archival pages also link to the pinned source on GitHub. The verification report records the Lean and mathlib versions, the build, and the axiom checks.

A.1. **Sheafiness.**

Definition A.1 (The rational-cover predicate)**.** The proofs are carried out on finite rational covers, so the formalisation uses the class `IsSheafy`. We chose this formulation because it is easier to work with for our purposes: it speaks directly about the completed rational localisations which occur in the proofs. The equivalences below show that this choice does not change the notion of sheafiness. The class is in the `ValuationSpectrum` namespace and is defined in `StructureSheaf.lean`. Here `RationalCoveringData A` consists of a rational datum for the base and finitely many data whose rational subsets cover it. The separate condition `C.IsRational` says that the numerator set in the base and in each member of the cover generates an open ideal. For a Tate ring this is equivalent to generating the unit ideal. Finally, `presheafValue D` is the completed rational localisation $A\langle T/s\rangle$ attached to D.

```
class IsSheafy (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [inst₁ : PlusSubring A]
    [inst₂ : IsHuberRing A] [T2Space A] [NonarchimedeanRing A]
    [letI : UniformSpace A :=
      IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A]
    [IsRingOfIntegralElements (A⁺)] : Prop where
  embedding : ∀ (C : RationalCoveringData A), C.IsRational →
    Topology.IsEmbedding (productRestrictionSub A C)
  gluing : ∀ (C : RationalCoveringData A), C.IsRational →
    ∀ (f : ∀ (D : ↑C.covers), presheafValue D.1),
    (∀ (D₁ D₂ : ↑C.covers) (D₃ : RationalLocData A)
      (h₃₁ : rationalOpen D₃.T D₃.s ⊆
        rationalOpen D₁.1.T D₁.1.s)
      (h₃₂ : rationalOpen D₃.T D₃.s ⊆
        rationalOpen D₂.1.T D₂.1.s),
      restrictionMap D₁.1 D₃ h₃₁ (f D₁) =
        restrictionMap D₂.1 D₃ h₃₂ (f D₂)) →
    ∃ x : presheafValue C.base, ∀ (D : ↑C.covers),
      restrictionMap C.base D.1 (C.hsubset D.1 D.2) x = f D
```

The two fields say that restriction to a finite rational cover is a topological embedding and that every matching family glues. Compatibility is written using common rational refinements; the exact-intersection theorem identifies this with the usual compatibility on pairwise intersections. Thus `IsSheafy A` is the topological sheaf condition on the rational basis.

Definition A.2 (Sections on arbitrary opens)**.** Let $V \subseteq \mathrm{Spa}(A, A^+)$ be open. The ring `limitSections V` consists of compatible sections over the rational subdomains contained in V, or equivalently their projective limit. If $W \subseteq V$, then `limitRestrict` is the map $\mathcal{O}_X(V) \to \mathcal{O}_X(W)$ obtained by reindexing this family. The corresponding all-open condition is `IsLimitSheaf`, from `SheafyPair.lean`.

The class `HasLocLiftPowerBounded` is used internally to construct the restriction maps between completed rational localisations. For an inclusion $R(D') \subseteq R(D)$, it says that the denominator of D becomes a unit on $R(D')$ and that the fractions t/s, for t in the numerator set of D, are power-bounded there. This class is supplied automatically for the complete Tate pairs considered here; it is not an additional hypothesis in our results.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
    [HasLocLiftPowerBounded A] : Prop where
  injective : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    {x y : ↑(limitSections V)},
    (∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
  glue : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
```

```
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    (s : ∀ i, ↑(limitSections (U i))),
    (∀ i j,
      limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
      limitRestrict (inf_le_right (a := U i) (b := U j)) (s j)) →
    ∃ x : ↑(limitSections V), ∀ i, limitRestrict (hle i) x = s i
  isEmbedding : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺))),
    Topology.IsEmbedding (limitRestrictProd hle)
```

The fields `injective` and `glue` are the usual sheaf axioms on arbitrary opens. The final field says that restriction identifies the sections over V homeomorphically with the subspace of $\prod_i \mathcal{O}(U_i)$ consisting of tuples whose restrictions agree. Together these conditions give the concrete form of Wedhorn's sheaf condition for topological rings.

Remark A.3 (Comparison with Mathlib)**.** For complete Tate pairs the rational-basis and all-open predicates are equivalent:

```
theorem isSheafy_iff_isLimitSheaf :
    IsSheafy A ↔ IsLimitSheaf A :=
  ⟨fun _ => isLimitSheaf_of_isSheafy,
    isSheafy_of_isLimitSheaf⟩
```

The forward implication is the sheaf-on-a-rational-basis theorem, and the converse is restriction to rational covers. For a bundled presheaf F, `F.IsSheafOfTopologicalRings` means that, for every topological commutative ring T, the presheaf $U \mapsto \mathrm{Hom}_{\mathrm{cont}}(T, F(U))$ is a sheaf of sets. This is Wedhorn's representable formulation [20, Remark 8.20]. For the structure presheaf the formalisation proves:

```
theorem structurePresheaf_isSheafOfTopologicalRings_iff :
    TopCat.Presheaf.IsSheafOfTopologicalRings
        (structurePresheaf A) ↔
      IsLimitSheaf A
```

The displayed theorem, together with `isSheafy_iff_isLimitSheaf`, gives explicitly

```
IsSheafy A ↔
  (structurePresheaf A).IsSheafOfTopologicalRings
```

Thus the bundled `IsSheafOfTopologicalRings` condition and the rational-cover condition used in the proofs are equivalent; neither is merely a consequence of the other.

There is also a direct comparison with Mathlib's categorical definition. For an arbitrary presheaf F of complete separated topological commutative rings, the general comparison theorem is

```
theorem isSheafOfTopologicalRings_iff_forgetToTopCommRingCat_isSheaf :
    F.IsSheafOfTopologicalRings ↔
      TopCat.Presheaf.IsSheaf
        (F ≫ CompleteTopCommRingCat.forgetToTopCommRingCat)
```

The functor in this statement forgets completeness and separatedness, but not the topology. The objects of `TopCommRingCat` are all topological commutative rings, so Mathlib's categorical condition tests exactly the presheaves $U \mapsto \mathrm{Hom}_{\mathrm{cont}}(T, F(U))$ appearing in Wedhorn's definition.

By contrast, the categorical condition with values left in `CompleteTopCommRingCat` tests only complete separated rings and is only a consequence of Wedhorn's condition.

Specialising the comparison to the structure presheaf gives the all-open equivalence. Combining it with the rational-basis theorem above gives the final comparison:

```
theorem isSheafy_iff_structurePresheaf_forgetToTopCommRingCat_isSheaf :
    IsSheafy A ↔
      TopCat.Presheaf.IsSheaf
        (structurePresheaf A ≫
          CompleteTopCommRingCat.forgetToTopCommRingCat)
```

Thus all three predicates are equivalent. We use `IsSheafy` because it is the convenient form for the arguments in this paper, not because it defines a weaker or stronger notion of sheafiness.

For a complete Tate ring the following definitions fix a ring of integral elements and then remove the choices of plus ring and completion. They are in `SheafyRing.lean`.

```
def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
  letI := Aplus.toPlusSubring
  haveI : IsRingOfIntegralElements (A⁺ : Subring A) := Aplus.2
  haveI : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
  IsLimitSheaf A

def IsSheafyComplete : Prop :=
  ∀ Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
  ∀ (P : PairOfDefinition A)
    (Bplus : RingOfIntegralElements (CompletionModel A P)),
    IsSheafyFor (CompletionModel A P) Bplus
```

The predicate `IsSheafyFor A Aplus` says that the Huber pair (A, A^+) is sheafy. For complete Tate rings, the chosen-plus-ring equivalence proves that the choice of integral elements does not affect the condition. The equivalences in `SheafyCompletionModel.lean` remove the choice of completion. Consequently `IsSheafyTateRing` is the analytic specialisation of Wedhorn's Definition 8.26 [20, Definition 8.26].

A.2. **Strong noetherianity.**

Definition A.4 (Strong noetherianity)**.** We use the following definitions. They are from `RestrictedPowerSeries.lean`.

```
def MvPowerSeries.IsRestricted {k : ℕ} {A : Type*}
    [CommRing A] [TopologicalSpace A]
    (f : MvPowerSeries (Fin k) A) : Prop :=
  Tendsto (fun s : Fin k →₀ ℕ => MvPowerSeries.coeff s f)
    cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
    [TopologicalSpace A] [NonarchimedeanRing A] : Prop where
  isNoetherianRing_restricted : ∀ k : ℕ,
    IsNoetherianRing (restrictedMvPowerSeriesSubring k A)
```

The cofinite convergence condition is the usual condition that the coefficients tend to zero, so `restrictedMvPowerSeriesSubring k A` represents $A\langle T_1, \ldots, T_k\rangle$. The second declaration therefore says that these rings are noetherian for every k. Since A is already complete, this is Wedhorn's definition of strong noetherianity [20, Proposition and Definition 6.36].

A.3. **The finite-jet example.**

Definition A.5 (The finite-jet ring)**.** We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over R, `RestrictedLaurent K` is $K\langle W, W^{-1}\rangle$, and `DualNumber R` is $R[Q]/(Q^2)$. The source is `FJP/Over/JetRings.lean`.

```
abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)

abbrev JetB : Type u :=
  DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

noncomputable def jetSupport : Subring (JetC K) where
  carrier := {f |
    qCoeff K 0 f ∈ nonnegSubring K ∧
    qCoeff K 1 f ∈ nonnegSubring K}
  zero_mem' := by
    constructor <;> rw [qCoeff_zero] <;> exact zero_mem _
  one_mem' := by
    constructor <;> rw [qCoeff_one]
    · rw [if_pos rfl]; exact one_mem _
    · rw [if_neg one_ne_zero]; exact zero_mem _
  add_mem' := fun {f g} hf hg => by
    constructor <;> rw [qCoeff_add]
    · exact add_mem hf.1 hg.1
    · exact add_mem hf.2 hg.2
  neg_mem' := fun {f} hf => by
    constructor <;> rw [qCoeff_neg]
    · exact neg_mem hf.1
    · exact neg_mem hf.2
  mul_mem' := fun {f g} hf hg => by
    constructor
    · rw [qCoeff_zero_mul]
      exact mul_mem hf.1 hg.1
    · rw [qCoeff_one_mul]
      exact add_mem (mul_mem hf.1 hg.2) (mul_mem hf.2 hg.1)

abbrev JetA : Type u := ↑(jetSupport K)
```

The predicate `nonnegSubring K` cuts out $K\langle W\rangle$ inside $K\langle W, W^{-1}\rangle$. Consequently the carrier condition says that the coefficients of Q^0 and Q^1 have no negative powers of W, while the coefficients of Q^n for $n \geq 2$ are unrestricted. Thus `JetA K` is

$$\left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) \;\middle|\; {\textstyle \substack{f_0, f_1 \in K\langle W\rangle \\ h \in L\langle Q\rangle}} \right\},$$

which is the coefficient description of A in equation (5) after inverting ϖ.

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in `JetB K` and `JetC K` comes from a unique element of `JetA K`. This identifies `JetA K` with $B \times_D C$. The sheaf argument produces the finite-rational-cover condition displayed above:

```
include ϖ hK₀ in
theorem isSheafy_JetA :
    ValuationSpectrum.IsSheafy (JetA K) where
  embedding := fun C hC =>
    productRestrictionSub_isEmbedding_JetA ϖ hK₀ C hC
  gluing := fun C hC f hcompat =>
    gluing_JetA ϖ hK₀ C hC f hcompat
```

Here ϖ is a `Uniformizer K`, and the remaining input is the noetherianity of `unitBall K`. This is the version of sheafiness proved directly for the example. The equivalences above give the all-open and choice-independent statements. The main theorems are collected in `SheafyEndpoints`, and include the following.

```
theorem finiteJet_isDomain : IsDomain (JetA K) :=
  inferInstance

theorem finiteJet_not_noetherian :
    ¬ IsNoetherianRing (JetA K) :=
  not_isNoetherianRing_JetA K

theorem finiteJet_isUniform_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    TopologicalRing.IsUniform (JetA K) :=
  finiteJet_isUniform K (Uniformizer.ofDVR K)

theorem finiteJet_not_stablyUniform_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    ¬ TopologicalRing.IsStablyUniform (JetA K) :=
  finiteJet_not_stablyUniform K (Uniformizer.ofDVR K)

theorem finiteJet_isSheafyFor_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyFor (JetA K) (finiteJetPlus K) :=
  finiteJet_isSheafyFor K (Uniformizer.ofDVR K)
    (isNoetherianRing_unitBall K)

theorem finiteJet_isSheafyTateRing_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyTateRing (JetA K) :=
  finiteJet_isSheafyTateRing K (Uniformizer.ofDVR K)
    (isNoetherianRing_unitBall K)
```

Here K is any complete nontrivially normed ultrametric field for which $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$ is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The theorem `finiteJet_isSheafyFor_of_dvr` proves that (A, A°) is sheafy, whilst `finiteJet_isSheafyTateRing_of_dvr` removes the choices of completion and ring of integral elements. The corresponding all-open statement is

```
TopCat.Presheaf.IsSheafOfTopologicalRings
    (ValuationSpectrum.structurePresheaf (JetA K)),
```

The corresponding Lean theorem proves this statement directly.

A.4. **Scope of the finite-jet formalisation.** The paper's A_0 is represented by the Gauss-norm unit ball. The theorem `isPowerBounded_JetA_iff` identifies this unit ball with the power-bounded elements, giving $A^\circ = A_0$. Completeness and the Tate structure are supplied by the corresponding instances. As in the weighted-parity development, the constant-series map from K is explicit and isometric but is not installed as an `Algebra K` instance.

For the bad chart, `chartEquiv` is an exported ring equivalence. The two continuity theorems make it a homeomorphism, and the canonical-map theorem identifies W and Q. It is not bundled as a `K-AlgEquiv`, which is why proposition 4.1 is stated for complete topological rings.

The consequences in section 7 are formalised over $K = F((t))$. This is less general than the arbitrary complete discretely valued field used in the paper, but taking $F = \mathbb{Q}$ is enough to answer the two existence questions. The declaration `finiteJet_witnessRing_quality` records that this concrete ring is a uniform sheafy Tate domain and is not noetherian. The theorem `finiteJet_problem28` records in one statement that $(W; \varpi)$ is rational, its rational subset and section ring are nonempty and nonzero, multiplication by Q^2 is a continuous strict injection with closed image, and Q^2 restricts to zero. The nonempty point is constructed from the norm valuation pulled back along `ev00`, the map

$$f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(0).$$

The separate declarations saying that multiplication by Q^2 is an isometry and that Q^2 restricts to zero are the inputs used below.

```
theorem finiteJet_problem28 :
    (chartDatum F).IsRational ∧
    (rationalOpen (chartDatum F).T (chartDatum F).s).Nonempty ∧
    Nontrivial (presheafValue (chartDatum F)) ∧
    Function.Injective
      (fun a : JetA F => scottishWitness F * a) ∧
    Continuous (fun a : JetA F => scottishWitness F * a) ∧
    IsStrictMap (fun a : JetA F => scottishWitness F * a) ∧
    IsClosed (Set.range
      fun a : JetA F => scottishWitness F * a) ∧
    (chartDatum F).canonicalMap (scottishWitness F) = 0
```

For Problem 24, Lean uses the module structure induced by the canonical map to the completed localisation and proves `finiteJet_not_flat_canonicalMap`. The proof is the one given in proposition 7.1: Q^2 is a non-zero-divisor of A, so flatness would make its image act injectively on the chart; that image is zero and the chart is nonzero. The ordinary ring localisation `Localization.Away` is always flat. The failure is a property of the completion defining `presheafValue`.

The conclusions of lemma 5.1 are formalised in the normed setting used for the example: E is one of B, C, D, T_E has its Gauss norm, $T_{E,0}$ is its unit ball, and the finite free terms have their sup norms. Positive-degree exactness is transported from the polynomial sequence by flat base change. Closedness in positive degree comes from exactness and continuity, whilst the degree-zero ideal is closed by the noetherian closed-ideal theorem. The closed-range theorem gives a constant $C_{q,E}$ such that every $y \in \mathrm{im}(d_{q,E})$ has a preimage x with $\|x\| \le C_{q,E}\|y\|$. This proves strictness, and multiplying such a preimage by a suitable power of ϖ gives the two displayed denominator inclusions.

The Lean proof of lemma 5.2 uses these inequalities directly. Its localisation proof obtains the topological embedding from a quotient-norm estimate, and proves openness of $C_\alpha \to D_\alpha$ from the norm-preserving coefficientwise section. The components of proposition 5.3 are therefore separate declarations. Finally, the development proves the concrete finite-jet transfer `isSheafy_JetA`: it pushes finite rational covers to B, C, D, glues there, and uses the localised Milnor row and a closed-range theorem.

The Koszul argument, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

A.5. **The weighted-parity example.** The second development uses the same restricted-series infrastructure. Its ambient ring is the countable restricted Tate algebra

$$K\langle W, U_1, U_2, \ldots \rangle.$$

The definitions `wpWeight`, `WPMem`, and `wpSupport` are the direct translation of (14)–(18). The relevant part of the definition is as follows.

```
def wpWeight (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : ℕ :=
  ∑ n ∈ t.support,
    if (t n).mod 2 = 1 ∧ n ≠ 0 then w n else 0

def WPMem (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : Prop :=
  wpWeight w t ≤ t 0

abbrev WPA : Type _ := ↑(wpSupport K w)

def idWeight : ℕ → ℕ := fun n => n

abbrev WPAid : Type _ := WPA K idWeight
```

The coordinate $t(0)$ is the exponent of W; the other coordinates are the exponents of the U_n. Thus `WPMem idWeight t` says exactly that $a \geq \omega(\nu)$, and `wpSupport` requires the coefficients outside this monoid to vanish. It follows that `WPAid K` is the ring $\mathcal{A}$ of (18). The maps `constA` and `norm_constA` give the isometric embedding of K as the constant series. This map is explicit in the development, although it is not bundled as an `Algebra K` instance.

Suppose that K is complete and nontrivially normed and that $\mathcal{O}[K]$ is a DVR. The principal exported statements are:

```
theorem weightedParity_isUniform_of_dvr :
    IsUniform (WPAid K) :=
  weightedParity_isUniform (Uniformizer.ofDVR K)

theorem weightedParity_isDomain :
    IsDomain (WPAid K) :=
  inferInstance

theorem weightedParity_not_noetherian :
    ¬ IsNoetherianRing (WPAid K) :=
  not_isNoetherianRing_WPA K idWeight idWeight_pos

theorem weightedParity_powerBounded_eq_unitBall
    (ϖ : Uniformizer K) :
    powerBoundedSubring (WPAid K) =
      (FiniteJet.unitBall (WPAid K) : Set (WPAid K)) :=
  powerBoundedSubring_eq_unitBall ϖ

theorem weightedParity_isSheafyComplete_of_dvr :
    IsSheafyComplete (WPAid K) :=
  weightedParity_isSheafyComplete (Uniformizer.ofDVR K)
```

```
    (FiniteJetOver.isNoetherianRing_unitBall K)

theorem weightedParity_chart_isDomain
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    IsDomain (presheafValue
      (chartDatum (w := idWeight) ϖ)) :=
  isDomain_chart ϖ hK₀

theorem weightedParity_chart_not_isUniform
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    ¬ IsUniform (presheafValue
      (chartDatum (w := idWeight) ϖ)) :=
  not_isUniform_chart idWeight_unbounded ϖ hK₀

theorem weightedParity_not_stablyUniform_of_dvr :
    ¬ IsStablyUniform (WPAid K) :=
  weightedParity_not_stablyUniform (Uniformizer.ofDVR K)
    (FiniteJetOver.isNoetherianRing_unitBall K)
```

Here `IsSheafyComplete` is the predicate defined above: it quantifies over every ring of integral elements. In particular it proves the sheaf condition for $(\mathcal{A}, \mathcal{A}^\circ)$. The all-open theorem also records the corresponding statement directly for the structure presheaf. The datum `chartDatum` is $(W; \varpi)$, and `chartDatum_isRational` checks that it defines the chart $|W| \leq |\varpi|$. The two chart theorems then say precisely that its section ring is a domain but is not uniform.

The proof routes are the ones used in section 8. Multiplicativity of the countable Gauss norm gives the domain and unit-ball statements. For rational data in $\mathcal{A}^{(N)}$, localisation is proved by taking the completed quotient of $\mathcal{A}^{(N)}$, applying its evaluation map to each coefficient in (23), and constructing the inverse by summing coefficients that tend to zero; it does not invoke the Koszul lemma. For sheafiness, one N is chosen for the base datum and all members of a finite cover. Strong noetherianity of $\mathcal{A}^{(N)}$ glues each coefficient, and the restriction embedding shows that the resulting coefficients tend to zero. The final topological embedding is obtained from the closed matching equaliser by the Tate closed-range theorem. For the bad chart, the quotient formed from $\mathcal{A}^{(0)} = K\langle W\rangle$ is $K\langle X\rangle$, and the coefficientwise ring embeds in formal multivariable power series over this domain. The formalisation also contains, under `WeightedParity.PerturbSetup`, a quantitative norm-unit-ball version of the standard perturbation result cited in lemma 8.5.

These declarations, including the final failure of stable uniformity, have been checked to use only the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

A.6. **Mathematical and code provenance.** The formal structure presheaf, rational-cover criterion, and the strongly-noetherian sheaf theorem follow Huber's construction [10, Theorem 2.2 and Lemmas 2.3–2.6] and Wedhorn's systematic formulation [20, Proposition and Definition 6.36, Remark 8.20, Definition 8.26, Theorem 8.28(b), Corollary 8.32, and Lemmas 8.33–8.34]. The passage between a fixed plus ring and all plus rings follows the standard rational-refinement argument recorded by Kedlaya [11, Lemma 1.6.8 and Remark 1.6.9]. Bambozzi–Kremnizer formulate the general construction in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, and Corollary 4.13]. The formalisation does not invoke this result. Positive-degree exactness is instead proved by flat base change from the polynomial sequence; closedness and the nonarchimedean open mapping theorem [13, Lemma 3.1] give constants bounding the norm of a preimage by a fixed multiple of the norm of its image.

Multiplication by a suitable power of ϖ then yields the bounded-denominator criterion used by Buzzard–Verberkmoes [4, Lemma 2].

The development is written in Lean 4 [14] on top of mathlib [15]. The formalisation-specific files live in AINTLIB [1]. AINTLIB includes code adapted from William Coram's restricted-power-series development [6]; its univariate Gauss-norm layer builds on Fabrizio Barroero's mathlib code [5]. It separately uses Bingyu Xia's multivariable power-series equivalences [8]. The vendored source headers and commit history retain the respective attributions. The finite-jet and weighted-parity definitions, localisation proofs, and sheaf endpoints discussed in this appendix are the AINTLIB-specific part of the development.

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